

04_FileMakerAPI.md

MonitorAI v1.5 FileMaker API Specification

Version: 1.5

Status: Baseline

1. Purpose

This document defines how the MonitorAI iPhone application communicates with FileMaker Server.

The API layer is responsible for:

- Authentication
 - Session creation
 - Capture record creation
 - Image upload
 - Metadata upload
 - Upload queue recovery
 - Session completion
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2. Backend

Backend System:

FileMaker Server 19

API:

FileMaker Data API

Transport:

HTTPS

Authentication:

Bearer Token

3. Core Principles

The application must never lose:

- Images
- Metadata
- Session data

All upload operations must be:

- Traceable
 - Retryable
 - Recoverable
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4. Security Requirements

Required:

HTTPS

TLS 1.2+

Bearer Token Authentication

Keychain Token Storage

Prohibited:

Hardcoded Credentials

Plain Text Password Storage

Patient Information in Debug Logs

Unencrypted Upload

5. Data API Layouts

Dedicated API layouts must be created in FileMaker.

Do not use clinical UI layouts for API communication.

Required API Layouts:

MonitorAI_API_Sessions

MonitorAI_API_Captures

MonitorAI_API_Metadata

MonitorAI_API_UploadQueue

MonitorAI_API_TrainingLabels

MonitorAI_API_SystemSettings

6. Authentication Flow

Login



Receive Token



Store Token in Keychain



Use Token for API Calls



Refresh / Re-login if expired



Logout when appropriate

7. Session Creation

Triggered:

After patient selection and before Position Calibration.

Purpose:

Create a session record.

Required Fields:

- SessionUUID
 - PatientUUID
 - SessionStartTime
 - DeviceModel
 - DeviceModelIdentifier
 - iOSVersion
 - AppVersion
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8. Position Calibration Upload

Position Calibration data is uploaded as part of session or first capture metadata.

Required Fields:

- CalibrationStartTime
 - CalibrationEndTime
 - CalibrationDurationSeconds
 - CalibrationOffsetX
 - CalibrationOffsetY
 - CalibrationQualityScore
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9. Capture Record Creation

One CaptureImages record must be created for each accepted image.

Expected:

11 final accepted capture records per completed session.

Required Fields:

- CaptureUUID
 - SessionUUID
 - CaptureStep
 - CaptureName
 - CaptureStartTime
 - CaptureSuccessTime
 - CaptureDurationSeconds
 - RetakeVersion
 - FinalAccepted
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10. Image Upload

Image files are uploaded to container fields.

Required Containers:

- ImageFile
- ThumbnailFile

Storage:

External Secure Storage

Upload Type:

Container Upload

11. Metadata Upload

Each accepted capture must have a CaptureMetadata record.

Required Metadata Groups:

- PTG Metadata
- Visibility Metadata
- Camera Metadata
- Lighting Metadata
- Image Quality Metadata
- Timing Metadata
- Voice Metadata
- Haptic Metadata

- Upload Metadata
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12. Upload Timing

For every upload, store:

- UploadStartTime
- UploadCompleteTime
- UploadDurationSeconds

Purpose:

Evaluate upload performance and backend reliability.

13. Offline Queue

If upload fails:

Store locally

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Create UploadQueue item

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Retry automatically

↓

Complete upload when connection returns

14. Retry Strategy

Retry schedule:

1 Minute

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5 Minutes



15 Minutes



1 Hour



Manual Retry

Retry count must be stored.

Last error must be stored.

15. Session Completion

Session may be marked complete only when:

- 11 final accepted images uploaded
 - All metadata uploaded
 - Upload queue cleared
 - Session status updated
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16. Error Handling

Authentication Error:

Re-login

Network Error:

Offline Queue

Upload Error:

Retry Queue

Server Error:

Store Locally

Retry Later

17. API Success Criteria

An API workflow is successful when:

- Session record created
 - 11 CaptureImages records created
 - 11 ImageFile uploads completed
 - 11 ThumbnailFile uploads completed
 - 11 CaptureMetadata records created
 - Session marked complete
 - No local queue remains
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18. Final Principle

FileMaker is not passive storage.

FileMaker is the operational and dataset repository for MonitorAI.

The API must preserve:

- Image integrity
- Metadata integrity
- Session traceability
- Future AI dataset value